Joint Radar-Communications Performance Bounds: Data versus Estimation Information Rates

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***Abstract*—We investigate methods of co-existence between radar and communications systems. Since historically each system views the other as a source of interference, the systems are traditionally isolated spectrally or spatially in order to prevent performance degradation. By considering a cooperative radar and communications signaling scheme, we derive achievable bounds on performance for a receiver that observes communications and radar return in the same frequency allocation. These inner bounds on performance will be derived in scenarios where the radar system has to perform target parameter estimation for multiple targets. Additionally, a novel inner bound that maximizes the Fisher information for radar parameter estimation is also derived. Bounds on performance of the joint system are measured in terms of data information rate for communications and radar estimation information rate parameterization for the radar.**

***Index Terms*—Radar-Communications Co-existence**

I. Introduction

There is an ever increasing demand for spectrum and given the limit on resources, communications and radar systems are increasingly encouraged to share bandwidth. This can cause inter-system interference that degrades the performance of both systems. The standard solution is to separate (temporally, spatially or spectrally) the radar and communications systems. In this paper, we consider a system that does not require this separation, and we construct performance bounds for such a joint radar-communications system.

In this paper, we refine and extend the performance bounds introduced in [1] and portions discussed in [2]. The bounds presented in [1] are extended to include target parameter estimation for multiple independent targets that are far apart. We also discuss the problems associated with deriving performance bounds when the targets are neither independent nor far apart. In this paper, two targets are independent if the radar tracking motion for both targets are completely decorrelated (independent). Finally, a new inner bound on performance, the optimal Fisher information bound, is also discussed in this paper.

It is worth noting that in our efforts presented here, we focus on the radar estimation performance rather than radar detection considered in [3]–[5]. To be more specific, in our work, we have focused on the estimation of a dynamic target parameter, time delay or target range, from the received target return and the performance of the radar system is measured in terms of the estimation rate. The work presented in [3] investigated the application of information theory to improve radar system performance. In those works, by using information theory, the *a posteriori* radar receiver is developed that maximizes the quantity of information (instead of output SNR), given by the *a posteriori* distribution of a target parameter. In [4], it is shown that radar performance can be improved by designing radar waveforms that maximize the mutual information between the target parameter and the received measurements.

Current research has investigated the benefits of using methods similar to cooperative sensing to solve the problem of radar and communications co-existence [6], [7]. Techniques such as interference mitigation [8], precoding (spatial separation) [9] or waveform shaping [10] allow both systems to share the spectrum. Radar systems using communications waveforms (OFDM or DSSS signals) as illumination signals have also been considered [11]–[13]. Similarly, communications systems using radar waveforms (linear frequency modulation (LFM) chirp) as modulation signals have been developed [14].

In this paper, we consider the joint radar-communications receiver to be able to simultaneously estimate the radar target parameters from the radar return and decode a communications signal. The communications relay will not be discussed in this paper. The inner bounds on performance are found by hypothesizing this idealized receiver and determining the bounding information rates.

This paper is organized as follows. In Section II, we present the channel and signal model for a joint radar-communications system that will be employed in this paper. We also introduce the idea of an observed signal with a predicted radar return signal suppressed. In Section III, we develop the radar estimation information rate, the metric used to measure radar system performance for multiple independent, far apart targets. We also discuss the problems with finding the estimation rate for multiple correlated, close together targets. In Section IV, we construct the optimal fisher information inner bounds on the performance of a joint radar-communications system. Finally, in Section V, we draw conclusions from the results obtained in this paper and discuss avenues for future research.

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II. Joint Radar-Communications Channel Model

In this section, we consider the joint radar-communications system complex baseband received signal, $z(t)$, for a multiple-access communications and radar return channel. We assume that the residual of the unpredicted radar return is modeled well by a Gaussian distribution and that target range is known up to some Gaussian random process variation which is within one over the bandwidth. We consider only the portion of time during which the radar return overlaps with the communications signal. We present a table of significant notation that will be employed in this paper in Table I.

TABLE I
Survey of Notation

Variable	Description
$\langle\cdot\rangle$	Expectation
$\|\cdot\|$	L2-norm or absolute value
$\theta, \boldsymbol{\theta}$	Estimation parameters
B	Full bandwidth of the system
$B_{\rm rms}$	Root-mean-squared radar bandwidth
$x(t)$	Unit-variance transmitted radar signal
$P_{\rm rad}$	Radar power
τ_m	Time delay to m^{th} target
$\tau_m^{(k)}$	k^{th} observation of delay for m^{th} target
$\tau_{m,\rm pre}$	Predicted time delay to m^{th} target
a_m	Complex combined antenna, cross-section, and propagation gain for m^{th} target
T	Radar pulse duration
N	Number of targets
δ	Radar duty factor
γ	Radar spectral shape parameter
$r(t)$	Unit-variance transmitted communications signal
$P_{\rm com}$	Total communications power
b	Combined antenna gain and communications propagation loss (amplitude)
$n(t)$	Receiver thermal noise
$\sigma^2_{\rm noise}$	Thermal noise power
k_B	Boltzmann constant
$T_{\rm temp}$	Absolute temperature
$n_{\rm int+n}$	Interference plus noise for communications receiver
$\sigma^2_{\tau,\rm proc}$	Variance of range fluctuation process
$B_{\rm com}$	Communications only sub-band
$B_{\rm rad}$	Radar only sub-band
$B_{\rm mix}$	Mixed radar and communications sub-band
$\rho_{\rm RO}$	Power spectral density used by radar only sub-band
$\rho_{\rm MU}$	Power spectral density used by mixed use sub-band
α	Bandwidth fraction ratio for sub-band splitting
β	Power fraction used by communications only sub-band
$R_{\rm est}$	Radar estimation rate
$R_{\rm CO}$	Communications rate for communications only sub-band
$R_{\rm MU}$	Communications rate for mixed use sub-band

A. Radar Return Signal Model

It is assumed that the received signal has been passed through a brick-wall filter matched to the bandwidth of interest, B. For N targets, the observed complex baseband [15] radar return $z(t)$ in the presence of a communications signal and noise is given by

$$z(t) = b\,\sqrt{P_{\rm com}}\,r(t) + \sqrt{P_{\rm rad}}\,\sum_{m=1}^{N} a_m\,x(t-\tau_m)\;+n(t)\,. \tag{1}$$

The zero-mean thermal noise $n(t)$ is drawn from a complex Gaussian distribution with variance $\sigma^2_{\rm noise} = k_B\,T_{\rm temp}\,B$.

As stated earlier, we have some knowledge of the target delay (or target range), up to some fluctuation in the return due to an underlying target random process. This fluctuation is modeled by a zero-mean Gaussian distribution $n_{\tau_{\rm m},\rm proc}$. During the kth observation, the delay for the mth target will be $\tau_m^{(k)} = \tau_{m,\rm pre}^{(k)} + n_{\tau_{\rm m},\rm proc}$, where $\tau_{m,\rm pre}$ is the predicted target delay. The variance of the range fluctuation process is $\sigma^2_{\tau_{\rm m},\rm proc} = \left\langle \|n_{\tau_{\rm m},\rm proc}\|^2 \right\rangle$.

B. Communications Signal with Predicted Radar Return Suppressed

In order to mitigate interference caused by the radar signal and improve the performance of the communications system, we employ a process called Successive Interference Cancellation (SIC) [1] at the receiver by generating a predicted radar return (from the predicted target range) and subtracting it from the received signal. This approach is only useful if the error in delay is smaller that $1/B$.

Utilizing the same derivation as in [1], for N targets, a communications receiver employing SIC will have an interference plus noise variance given by [1], [2]

$$\sigma^2_{\rm int+n} = P_{\rm rad}\left(\sum_{m=1}^{N}\|a_m\|^2\,(2\pi)^2\,B^2_{\rm rms}\,\sigma^2_{\tau_{\rm m},\rm proc}\right) + \sigma^2_{\rm noise}, \tag{2}$$

where f is frequency, $X(f)$ is the frequency response of $x(t)$, and $B_{\rm rms}$ comes from employing Parseval's theorem [15]. $B_{\rm rms}$ is extracted from bandwidth B as $\gamma^2\,B^2 = (2\pi)^2\,B^2_{\rm rms}$, where the value γ is the scaling constant between B and $B_{\rm rms}$ times 2π [1].

III. Radar Estimation Information Rate

In this section, we present a novel parameterization of the radar in terms of information rate. If radar illumination on a target can be viewed as the target unwillingly communicating information on its parameters (range, cross-section etc.), the radar channel can be characterized by a uncooperative communications channel and the estimation rate can be viewed as a mutual information between the radar return and the target.

A. Multiple Independent, Far Apart Targets

The radar estimation rate for multiple targets can be easily constructed if the targets are far apart in range and if the targets have independent range fluctuation process noise (tracking motion of the targets are independent of each other). In such a scenario, the radar channel for each target can be viewed as an independent uncooperative communications channel and the total estimation rate will be the sum of each radar channel's estimation rate. This is analogous to the case for multiple independent communications channels [16].

Motivated by the mutual information rate (or radar estimation rate) in terms of estimation entropy, random process entropy of the radar and bits per pulse repetition interval $T_{\rm pri} = T/\delta$, the radar estimation information rate for multiple

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independent radar channels (multiple independent targets) is bounded by

$$R_{\rm est} \leq \sum_m \frac{h_{\tau_{\rm m},{\rm rr}} - h_{\tau_{\rm m},{\rm est}}}{T_{\rm pri}}\,, \tag{3}$$

where $h_{\tau_{\rm m},{\rm rr}}$ is the received signal entropy and $h_{\tau_{\rm m},{\rm est}}$ is the estimation entropy.

Noting that the received signal entropy (assuming both process uncertainty and estimation uncertainty are Gaussian), is given by [15], [16]

$$h_{\tau_{\rm m},{\rm rr}} = \frac{1}{2}\,\log_2\left[2\,\pi\,e\,(\sigma^2_{\tau_{\rm m},{\rm proc}} + \sigma^2_{\tau_{\rm m},{\rm est}})\right]\,, \tag{4}$$

and that the estimation entropy is given by [1], [2]

$$h_{\tau_{\rm m},{\rm est}} = \frac{1}{2}\,\log_2\left[2\,\pi\,e\,\sigma^2_{\tau_{\rm m},{\rm est}}\right]\,, \tag{5}$$

we see that the radar estimation information rate is bounded explicitly as [1], [2]

$$R_{\rm est} \leq \frac{\delta}{2\,T}\,\sum_m \log_2\left[1 + \frac{\sigma^2_{\tau_m,{\rm proc}}}{\sigma^2_{\tau_{\rm m},{\rm est}}}\right]\,, \tag{6}$$

where $\sigma^2_{\tau_{\rm m},{\rm est}} = \frac{k_B\,T_{\rm temp}}{2\gamma^2\,B\,(TB)\,\|a_m\|^2\,P_{\rm rad}}$ is the Cramér Rao lower bound for time-delay estimation [17]. It is worth noting, that by employing this estimation entropy in the rate bound, it is assumed that the estimator achieves the Cramér-Rao performance. If the error variance is larger, then the rate bound is lowered.

From Equation (6), we see that a decrease of mean squared estimation error (Cramér Rao lower bound) will result in an increase in radar estimation rate. It should be noted, however, that a target with a large range fluctuation process variance can also increase the radar estimation rate. Similar to the communications case, an increase in the radar estimation rate essentially implies that an increased amount of information is gained from radar illumination through the radar channel.

B. Multiple Correlated, Close Together Targets

For multiple targets that are close together and have correlated range fluctuation process noise (tracking motion of the targets are correlated to some degree), deriving the radar estimation rate becomes much harder. In the previous section, the estimation rate was easily constructed because we assumed that the Cramér-Rao lower bound for time delay estimation and the range fluctuation process noise for a target was independent of the other targets, which meant that the mutual information between the radar target and the radar return was independent for each target and the total estimation rate was the sum of the individual mutual information over all targets.

However, for two targets that are close together (even if they are resolvable), the Cramér-Rao lower bound for time delay estimation for one of those targets is dependent of the location of the other [18]. Additionally, if two targets are correlated (tracking motion of the targets are correlated), then they have dependent range fluctuation process noise. Hence, in scenarios where the targets are correlated and close together, the mutual information between the radar target and the radar return are no longer independent for each target and the total estimation rate is not equal to the sum of the mutual information for each target. When targets are correlated and close together, new methods have to be implemented to derive the total estimation rate, which are beyond the scope of this paper.

IV. Inner Bounds on Joint Radar Communications System Performance

In this section, we derive inner bounds on the performance of the joint radar-communications system. As mentioned earlier, performance is measured in data information rate for the communications system and estimation information rate for the radar system. As stated earlier, we consider N radar targets with the $m^{\rm th}$ target having delay τ_m and gain-propagation-cross-section product a_m. To simplify the discussion, we assume that all targets are far apart and that the range fluctuation process noise is independent for each target (radar tracking motion is independent for each target).

A. Isolated Sub-Band Inner Bound

In this section, we derive an inner bound by considering a scenario in which we partition the total bandwidth into two sub-bands, one for radar only and the other for communications, which is the standard solution. Each system functions without any interference in their respective sub-band [1].

The bandwidth will be split between the two sub-bands according to some α such that,

$$B = B_{\rm rad} + B_{\rm com}\,,\; B_{\rm com} = \alpha\,B\,,\; B_{\rm rad} = (1-\alpha)\,B\,.$$

The corresponding communications rate (for the communications only sub-band) is given by

$$\begin{aligned} R_{\rm com} &\leq B_{\rm com}\log_2\left[1 + \frac{\|b\|^2\,P_{\rm com}}{k_B\,T_{\rm temp}\,B_{\rm com}}\right] \\ &= \alpha\,B\log_2\left[1 + \frac{\|b\|^2\,P_{\rm com}}{k_B\,T_{\rm temp}\,\alpha\,B}\right]\,. \end{aligned} \tag{7}$$

and the corresponding radar estimation rate is given by

$$\begin{aligned} R_{\rm est} &\leq \frac{\delta}{2\,T}\sum_m \log_2\left[1 + \frac{2\sigma^2_{\tau_m,{\rm proc}}\,\gamma^2\,B^2_{\rm rad}\,T\,\|a_m\|^2\,P_{\rm rad}}{k_B\,T_{\rm temp}}\right] \\ &= \frac{\delta}{2\,T}\sum_m \\ &\qquad \log_2\left[1 + \frac{2\sigma^2_{\tau_m,{\rm proc}}\,\gamma^2\,(1-\alpha)^2 B^2\,T\,\|a_m\|^2\,P_{\rm rad}}{k_B\,T_{\rm temp}}\right]\,. \end{aligned} \tag{8}$$

B. Successive Interference Cancellation (SIC) Inner Bound

We discuss the technique of SIC that was mentioned in Section II-B. After suppressing the radar return, the receiver then decodes and removes the communications signal from the observed waveform to obtain a radar return signal free of communications interference. The resulting inner bound on performance is called the SIC bound [1].

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If $R_{\rm est} \approx 0$, it is as if the radar interference is not present and the communications system can operate at a data rate determined by the isolated communications bound,

$$R_{\rm com} \leq B \, \log_2\!\left(1 + \frac{\|b\|^2 \, P_{\rm com}}{k_B \, T_{\rm temp} \, B}\right) . \tag{9}$$

If $R_{\rm com}$ is sufficiently low for a given transmit power, then the receiver can decode and remove the communications signal from the observed waveform, leaving just the radar return. Thus, radar parameter estimation can be done without corruption from any outside interference. This implies that from the communications receiver's perspective, it observes interference plus noise as described by Equation (2),

$$R_{\rm com} \leq B \log_2\!\left[1 + \frac{\|b\|^2 \, P_{com}}{P_{\rm rad} \sum_{m=1}^{N} \|a_m\|^2 \, \gamma^2 \, B^2 \, \sigma^2_{\tau_{\rm m},{\rm proc}} + k_B \, T_{\rm temp} \, B}\right] , \tag{10}$$

In this regime, the corresponding estimation rate bound $R_{\rm est}$ is given by Equation (6). The SIC inner bound is given by the convex hull [19] between points given by Equations (9), (10) and (6).

C. Communications Water-filling Bound

In this section, we consider a scenario in which the total bandwidth is split into two sub-bands, one sub-band for communications only and the other sub-band for both radar and communications. It is not necessary that the sub-bands will be of equal bandwidth. We use water-filling to distribute the total communications power between the two sub-bands [1]. Water-filling optimizes the power and rate allocation between multiple channels [15], [16]. The mixed use channel operates at the SIC rate vertex defined by Equations (6) and (10).

Given some bandwidth separation α,

$$B = B_{\rm com} + B_{\rm mix} \, , \, B_{\rm com} = \alpha \, B \, , \, B_{\rm mix} = (1-\alpha) \, B \, , \tag{11}$$

then we optimize the power utilization, β, between sub-bands,

$$P_{\rm com} = P_{\rm CO} + P_{\rm MU} \, , \, P_{\rm CO} = \beta \, P_{\rm com} \, , \, P_{\rm MU} = (1-\beta) \, P_{\rm com} \, . \tag{12}$$

There are two effective channels

$$\mu_{\rm com} = \frac{b^2}{k_B \, T_{\rm temp} \, B_{\rm com}} \, , \, \mu_{\rm mix} = \frac{b^2}{\sigma^2_{\rm int+n}} \, , \tag{13}$$

$$\sigma^2_{\rm int+n} = \sum_{m=1}^{N} \|a_m\|^2 \, P_{\rm rad} \, \gamma^2 \, B^2_{\rm mix} \, \sigma^2_{\tau_{\rm m},{\rm proc}} + k_B \, T_{\rm temp} \, B_{\rm mix} \, . \tag{14}$$

The first for the communications only channel, and the second for the mixed use channel. We apply the water-filling result derived in [1] and see that the optimal power distribution (β) between the two sub-channels is given by:

$$\beta = \alpha + \frac{1}{P_{\rm com}} \left(\frac{\alpha - 1}{\mu_{\rm com}} + \frac{\alpha}{\mu_{\rm mix}} \right) ; \quad \text{when } \; P_{\rm com} \geq \frac{\alpha}{(1-\alpha) \, \mu_{\rm mix}} - \frac{1}{\mu_{\rm com}} \, . \tag{15}$$

The resulting communications rate bound in the communications-only sub-band is given by

$$R_{\rm CO} \leq \alpha \, B \log_2 \left[1 + \frac{\beta \, P_{\rm com} \, b^2}{k_B \, T_{\rm temp} \, \alpha \, B}\right] . \tag{16}$$

The mixed use communications rate inner bound is given by

$$R_{\rm MU} \leq (1-\alpha) \, B \, \log_2 \left[1 + \frac{b^2 \, (1-\beta) \, P_{\rm com}}{\sigma^2_{\rm int+n}}\right] , \tag{17}$$

where $\sigma^2_{\rm int+n}$ is given by Equation (14). The corresponding radar estimation rate inner bound is then given by

$$R_{\rm est} \leq \frac{\delta}{2 \, T} \sum_{m=1}^{N} \log_2 \left(1 + \frac{2 \sigma^2_{\tau_m,{\rm proc}} \gamma^2 \, B^2_{\rm mix} \, T \, \|a_m\|^2 \, P_{\rm rad}}{k_B \, T_{\rm temp}}\right) . \tag{18}$$

D. Optimal Fisher Information Inner Bound

In this section, we construct a new inner rate bound by splitting the total bandwidth into two sub-bands and distributing the radar power (or power spectral density) between the two sub-bands in a way that minimizes the Cramér-Rao lower bound (or maximizes the Fisher information) for time-delay estimation.

The bandwidth will be split between the two sub-bands according to some α such that,

$$B = B_{\rm rad} + B_{\rm mix} \, , \, B_{\rm rad} = \alpha \, B \, , \, B_{\rm mix} = (1-\alpha) \, B \, , \tag{19}$$

and we will optimize the power spectral densities, $\rho_{\rm RO}$ and $\rho_{\rm MU}$, utilized by the radar only and mixed use sub-bands respectively, to maximize the Fisher information, where

$$P_{\rm rad} = P_{\rm RO} + P_{\rm MU} \tag{20}$$

$$P_{\rm RO} = \alpha \, B \, \rho_{\rm RO} \, , \, P_{\rm MU} = (1-\alpha) B \, \rho_{\rm MU} \, . \tag{21}$$

Now, consider a radar signal $x(t)$ with bandwidth B, whose frequency spectrum $X(f)$ is centered around B_O. We assume that $X(f)$ is spectrally flat. We will now partition the frequency spectrum into two spectrally flat portions, $X_{\rm RO}(f)$ and $X_{\rm MU}(f)$ with bandwidths $\alpha \, B$ and $(1-\alpha)B$ respectively, thereby creating two new signals, $x_{\rm RO}(t)$ and $x_{\rm MU}(t)$ that will be transmitted in the radar only sub-band and mixed use sub-band respectively. Thus, after transmission, the radar receiver will observe the following return signal for N targets,

$$\begin{aligned} z(t) &= \sum_{m=1}^{N} a_m \sqrt{P_{\rm RO}} \, x_{\rm RO}(t - \tau_m) \\ &\quad + a_m \sqrt{P_{\rm MU}} \, x_{\rm MU}(t - \tau_m) + b \sqrt{P_{\rm com}} r(t) + n(t) \\ &= \sum_{m=1}^{N} a_m \, x(t - \tau_m) + b \, r(t) + n(t) \, , \end{aligned} \tag{22}$$

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where $x(t-\tau_m) = \sqrt{P_{\rm RO}}\, x_{\rm RO}(t-\tau_m) + \sqrt{P_{\rm MU}}\, x_{\rm MU}(t-\tau_m)$. Using this return signal, we derive the Cramér-Rao lower bound on the variance for time-delay estimation. Let $\theta = \tau$ be the parameter to be estimated. From equation (22), we see that $z(t) \sim \mathcal{CN}(\sum_{m=1}^{N} a_m\, x(t-\tau_m) + b\,\sqrt{P_{\rm com}}\, r(t), \sigma^2)$ and has the following probability density function,

$$p(z(t);\theta) = \frac{1}{\pi\sigma^2} \mathrm{e}^{-\frac{\|z(t)-\sum_{m=1}^{N} a_m\, x(t-\tau_m) - b\,\sqrt{P_{\rm com}}\, r(t)\|^2}{\sigma^2}} . \quad (23)$$

Now, the Fisher information for this estimation problem for the $m^{\rm th}$ target, J_m, is given by [2]

$$J_m = \left\langle \left\| \frac{n^*(t)\, a_m\, x'(t-\tau_m)}{\sigma^2} + c.c. \right\|^2 \right\rangle .$$

where c.c. stands for complex conjugate term and $x'(t-\tau_m) = \frac{\partial}{\partial \tau_m} x(t-\tau_m)$. On expanding and simplifying, we see that

$$J_m = \frac{\left(\begin{array}{l} 2\|a_m\|^2 \left\langle \left\| \sqrt{P_{\rm RO}}\, x'_{\rm RO}(t-\tau_m) \right. \right. \\ \left. \left. \quad + \sqrt{P_{\rm MU}}\, x'_{\rm MU}(t-\tau_m) \right\|^2 \right\rangle \end{array} \right)}{\sigma^2} . \quad (24)$$

By expanding and converting to frequency domain and applying Parseval's Theorem, the orthogonality of $X_{\rm RO}(f)$ and $X_{\rm MU}(f)$ and Equation (21), for spectrally flat $X_{\rm RO}(f)$ and $X_{\rm MU}(f)$, we get [2]

$$\begin{aligned} J_m = {} & \frac{8\pi^2\, \|a_m\|^2 \alpha\, TB\, \rho_{\rm RO}}{3k_B T_{\rm temp} B} \\ & \cdot \left[\left(B_O - \frac{B}{2} + \alpha B \right)^3 - \left(B_O - \frac{B}{2} \right)^3 \right] \\ & + \frac{8\pi^2\, \|a_m\|^2 (1-\alpha)\, TB\, \rho_{\rm MU}}{3k_B T_{\rm temp} B} \\ & \cdot \left[\left(B_O + \frac{B}{2} \right)^3 - \left(B_O - \frac{B}{2} + \alpha B \right)^3 \right] . \end{aligned} \quad (25)$$

We consider B_O to be a free parameter chosen such that the Fisher information for time-delay estimation, given by Equation (25), and the reduced Fisher information [15], [20] for time-delay estimation are equal. The resultant value for B_O is given by [2]

$$B_O = \frac{\alpha\, B\, (1-\alpha)[\rho_{\rm MU}(\alpha-1) + \rho_{\rm RO}\alpha]}{2(\rho_{\rm MU}(\alpha-1)^2 + \rho_{\rm RO}\alpha^2)} . \quad (26)$$

Applying Equations (26) and (20) in Equation (25) separately and using the first and second derivative tests, we can get the values of $\rho_{\rm RO}$ and $\rho_{\rm MU}$ that maximize the Fisher information [2]. The resultant estimation rate bound for the radar system in both sub-bands is given by

$$R_{\rm est} \le \frac{\delta}{2\,T} \sum_{m=1}^{N} \log_2\left[1 + \sigma^2_{\rm proc} J_m\right] , \quad (27)$$

where J_m is given by Equation (25) and related to the variance of the time-delay estimation by $\sigma^2_{\tau_{\rm m},{\rm est}} = J_m^{-1}$. The corresponding communications rate bound in the mixed use channel is

$$R_{\rm MU} \le (1-\alpha)\, B\, \log_2\left[1 + \frac{\|b\|^2\, P_{\rm com}}{\sigma^2_{\rm int+n}}\right] \quad (28)$$

$$\begin{aligned} \sigma^2_{\rm int+n} = {} & \sum_{m=1}^{N} \|a_m\|^2 (1-\alpha) B\, \rho_{\rm MU}\, \gamma^2\, (1-\alpha)^2 B^2 \sigma^2_{\rm proc} \\ & + (1-\alpha) k_B\, T_{\rm temp} B . \end{aligned}$$

We expect the resulting inner bound to have end points given by Equations (6) and (10) (SIC vertex) when $\alpha = 0$ and by Equation (6) when $\alpha = 1$.

E. Examples

In Figure 1, we display an example of the inner bounds on performance. The values of parameter sets #1 - #5 used in the example are displayed in Table II. Parameter sets #1 - #4 are used to generate the optimal Fisher information inner bounds shown in Figure 1a while parameter set #5 is used to generate the performance bounds shown in solid lines and parameter set #1 to generate the performance bounds shown in dashed lines in Figure 1b. The parameters not mentioned in sets #2 - #5 have the same values as in set #1. Only one target is present in Figure 1a while there are three targets present in Figure 1b. It is assumed that the communications signal is received through an antenna sidelobe, so that the radar and communications receive gain are not identical.

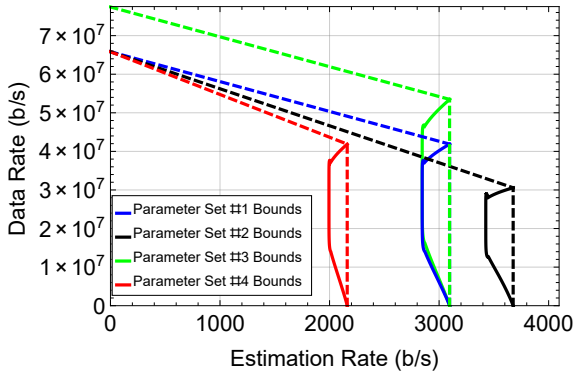
In Figure 1a, the SIC bound is indicated by dashed lines and the optimal Fisher bound by solid lines for a particular parameter set. We see that the end points of the optimal Fisher bound are as expected. In the example, as the radar power, $P_{\rm rad}$, is increased, as seen in parameter set #2, the optimal Fisher information bound shrinks (SIC vertex is lowered) and moves towards the right (radar estimation rate increases). Additionally, we see that as the communications power, $P_{\rm com}$, is increased, as seen in parameter set #3, the optimal Fisher information bound increases in height (SIC vertex is raised). Finally, we see that as the time-bandwidth product, TB, is increased, as seen in parameter set #4, the optimal Fisher information bound moves towards the left (radar estimation rate decreases.)

In Figure 1b, the isolated sub-band bound is indicated by the brown line. The SIC bound is indicated by green. The water-filling bound is indicated by the blue line. The optimal Fisher information bound is indicated by the black line. From Figure 1b, we see that under the simple assumptions about the multiple radar targets, the total estimation rate, which is the sum of each individual target estimation rate, increases as the number of targets increase which is what we expected.

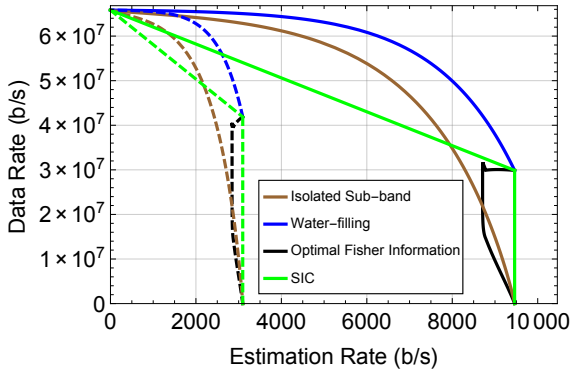
V. Conclusion

In this paper, using a previously developed estimation information rate for time-delay estimation, inner bounds on performance such as the isolated sub-band bound, the SIC inner bound, the water-filling bound and the optimal Fisher

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(a) Optimal Fisher Information Bounds for Parameter Sets #1-#4.



(b) Data Rate and Estimation Rate Bounds for 3 Targets

Fig. 1. Data Rate and Estimation Rate Bounds for Parameters in Table II

TABLE II
PARAMETERS FOR EXAMPLE PERFORMANCE BOUND

Parameter #1	**Value**
Bandwidth	5 MHz
Center Frequency	3 GHz
Temperature	1000 K
Communications Range	10 km
Communications Power	100 W
Communications Antenna Gain	0 dBi
Communications Receiver Side-lobe Gain	10 dBi
Radar Target Range	100 km
Radar Antenna Gain	30 dBi
Radar Power	100 kW
Target Cross Section	10 m^2
Target Process Standard Deviation	100 m
Time-Bandwidth Product	100
Radar Duty Factor	0.01
Parameter #2	**Value**
Radar Power	500 kW
Parameter #3	**Value**
Communications Power	500 W
Parameter #4	**Value**
Time-Bandwidth Product	200
Parameter #5	**Value**
Radar Target Range #2	80 km
Radar Target Range #3	120 km
Target Process Standard Deviation #2	130 m
Target Process Standard Deviation #3	90 m

information inner bound were extended to include target parameter estimation for multiple independent, far apart targets. Also, a discussion was provided addressing the the problems associated with deriving performance bounds when the targets are correlated or close together. Finally, a new performance bound, the optimal Fisher information bound, was derived and observed for different parameter sets.

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